

Random Forest — Non-Temporal Statistical Baseline

Per-model deep dive · the control condition for every temporal model

Result: 0.493 accuracy · 0.482 weighted F1 · WER 0.507 · fold SD ± 0.077

Notebook: RandomForest.ipynb (notebook 01)

1 Theory

A Random Forest is an ensemble of decision trees, each trained on a bootstrap sample of the data and a random subset of features, voting on the final class. It is a strong, low-variance classifier for tabular data of modest size. Crucially for this study it is **non-temporal**: it sees one fixed-length feature vector per video and has no notion of frame order. That makes it the deliberate **control condition** — the score a temporal model must beat to justify modelling time at all.

2 Method of execution

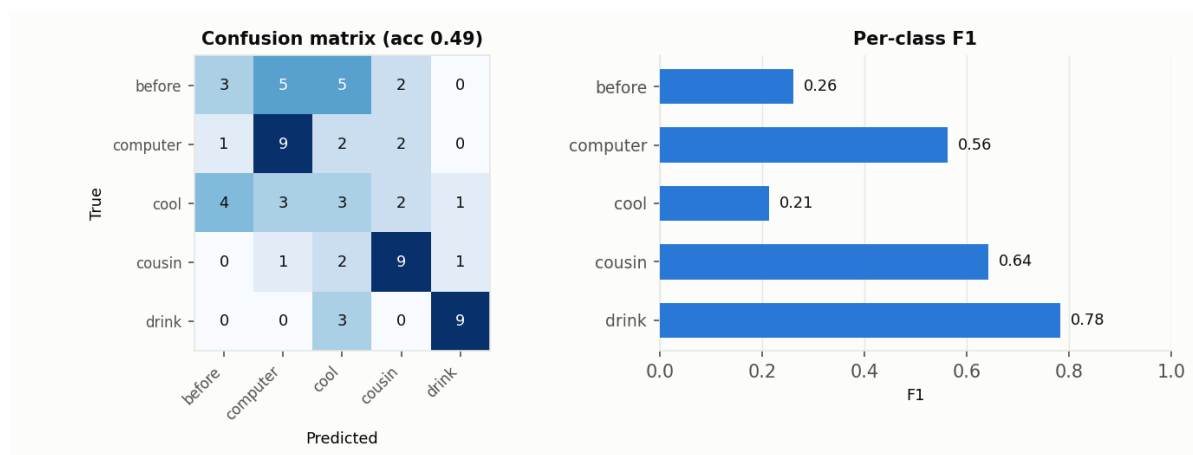
Each video is a sequence of 1,629-D MediaPipe landmark frames. The notebook collapses that sequence into a single vector by taking four statistics across the time axis — mean, standard deviation, minimum and maximum of every one of the 1,629 dimensions — giving a **6,516-D** summary. This is standardized and fed to a class-balanced forest of 300 trees.

```
stats = [frames.mean(0), frames.std(0), frames.min(0), frames.max(0)] # 4 × 1629
vector = concatenate(stats) # 6516-D
RandomForestClassifier(n_estimators=300, class_weight='balanced')
```

The key modelling decision (and deliberate weakness) is the pooling step: **mean/std/min/max destroy the order of frames**. A sign performed forwards and the same poses shuffled in time produce an identical feature vector. That is exactly the information a sign carries, thrown away on purpose to establish the baseline.

3 Results

Metric	Value (5-fold CV)
Accuracy	0.493
Weighted precision	0.490
Weighted recall	0.493
Weighted F1	0.482
Word Error Rate (WER)	0.507
Fold accuracies	0.36, 0.57, 0.46, 0.54, 0.54
Cross-val SD	± 0.077
Distinct classes predicted	5 of 5
Majority baseline	0.224



Random Forest confusion matrix and per-class F1 under 5-fold CV.

4 Why these results

At 0.493 the forest sits mid-table — above the HMM, BiLSTM and ST-GCN, below the Transformer and DTW. It clears the baseline comfortably because the *average* and *spread* of hand and body position already separate these five words to a useful degree, and forests are robust when data is scarce. Its low fold spread confirms it is a stable estimator. But it has a hard ceiling: any two signs that share pose statistics and differ only in movement order are indistinguishable to it — no amount of extra trees fixes that.

What makes it stronger / weaker

Stronger: robust on tiny data, fast, no temporal assumptions to violate, naturally handles the high-dimensional vector. **Weaker:** temporally blind by construction — the ceiling is set by how much signs happen to differ in static statistics. It is a yardstick, not a candidate for the final system.

5 Role in the hybrid

Random Forest is **not** a component of the recommended hybrid — its value is diagnostic. Because it captures only static, order-free information, the gap between it and a temporal model measures how much a given architecture actually extracts from motion. That the trained two-stream hybrid clears it by ~15 points is direct evidence the hybrid is using temporal structure, not just average pose.